

## 4.3 Alternate functions

Table 13. Alternate functions

| Port   |      | AF0               | AF1               | AF2                   | AF3                              | AF4                                                   | AF5                                             | AF6                                          | AF7                        | AF8                          | AF9                             | AF10     | AF11                       | AF12 | AF13      | AF14                   | AF15     |
|--------|------|-------------------|-------------------|-----------------------|----------------------------------|-------------------------------------------------------|-------------------------------------------------|----------------------------------------------|----------------------------|------------------------------|---------------------------------|----------|----------------------------|------|-----------|------------------------|----------|
|        |      | SYS               | LPTIM1/<br>TIM1/2 | I3C1/<br>TIM1/8/12/15 | I3C1/LPTIM1/<br>LPUART1/<br>TIM8 | I2C1/I3C1/<br>LPTIM1/SPI1/<br>I2S1/TIM15/<br>USART1/2 | I3C1/LPTIM1/<br>SPI1/I2S1/<br>SPI2/<br>I2S2/SYS | FDCAN2/<br>SPI1/<br>I2S1/SPI2/<br>I2S2/UART4 | SPI2/I2S2/<br>USART1/2     | I2C1/<br>LPUART1/<br>UART4/5 | FDCAN1/2/<br>I2C1/<br>SPI2/I2S2 | CRS      | USART1                     | -    | TIM8/USB_ | COMP1/2/<br>TIM2/UART5 | SYS      |
| Port A | PA0  | -                 | TIM2_CH1          | -                     | TIM8_ETR                         | TIM15_BKIN                                            | SPI2_RDY                                        | SPI1_RDY                                     | USART2_CTS<br>/ USART2_NSS | UART4_TX                     | SPI2_NSS/<br>I2S2_WS            | -        | -                          | -    | -         | TIM2_ETR               | EVENTOUT |
|        | PA1  | -                 | TIM2_CH2          | -                     | TIM8_BKIN                        | TIM15_CH1N                                            | LPTIM1_IN1                                      | -                                            | USART2_RTS                 | UART4_RX                     | -                               | -        | -                          | -    | -         | -                      | EVENTOUT |
|        | PA2  | -                 | TIM2_CH3          | -                     | LPUART1_RX                       | TIM15_CH1                                             | LPTIM1_IN2                                      | -                                            | USART2_TX                  | -                            | -                               | -        | -                          | -    | -         | COMP2_OUT              | EVENTOUT |
|        | PA3  | -                 | TIM2_CH4          | -                     | LPUART1_TX                       | TIM15_CH2                                             | SPI2_NSS/<br>I2S2_WS                            | SPI2_MOSI/<br>I2S2_SDO                       | USART2_RX                  | -                            | -                               | -        | -                          | -    | -         | COMP1_OUT              | EVENTOUT |
|        | PA4  | -                 | TIM1_BKIN2        | -                     | -                                | SPI1_MOSI/<br>I2S1_SDO                                | SPI1_NSS/<br>I2S1_WS                            | SPI2_NSS/<br>I2S2_WS                         | USART2_CK                  | -                            | -                               | -        | -                          | -    | -         | -                      | EVENTOUT |
|        | PA5  | -                 | TIM2_CH1          | TIM1_CH3              | TIM8_CH1N                        | -                                                     | SPI1_SCK/<br>I2S1_CK                            | SPI2_SCK/<br>I2S2_CK                         | USART1_CTS<br>/ USART1_NSS | -                            | -                               | -        | -                          | -    | -         | TIM2_ETR               | EVENTOUT |
|        | PA6  | -                 | TIM1_BKIN         | -                     | TIM8_BKIN                        | -                                                     | SPI1_MISO/<br>I2S1_SDI                          | -                                            | USART1_TX                  | LPUART1_RT<br>S              | -                               | -        | -                          | -    | -         | -                      | EVENTOUT |
|        | PA7  | -                 | TIM1_CH1N         | -                     | TIM8_CH1N                        | TIM15_CH1                                             | SPI1_MOSI/<br>I2S1_SDO                          | SPI2_MISO/<br>I2S2_SDI                       | USART1_RX                  | LPUART1_CT<br>S              | -                               | -        | -                          | -    | -         | COMP2_OUT              | EVENTOUT |
|        | PA8  | MCO1              | TIM1_CH1          | I3C1_SDA              | TIM8_BKIN2                       | TIM15_CH2                                             | SPI1_RDY                                        | SPI2_MOSI/<br>I2S2_SDO                       | USART1_CK                  | -                            | I2C1_SCL                        | -        | -                          | -    | USB_SOF   | TIM2_CH4               | EVENTOUT |
|        | PA9  | MCO2              | TIM1_CH2          | I3C1_SCL              | LPUART1_TX                       | TIM15_CH1N                                            | SPI2_SCK/<br>I2S2_CK                            | -                                            | USART1_TX                  | -                            | I2C1_SMBA                       | -        | -                          | -    | TIM8_CH2N | -                      | EVENTOUT |
|        | PA10 | -                 | TIM1_CH3          | -                     | LPUART1_RX                       | -                                                     | -                                               | -                                            | USART1_RX                  | -                            | FDCAN2_TX                       | -        | -                          | -    | -         | -                      | EVENTOUT |
|        | PA11 | -                 | TIM1_CH4          | -                     | LPUART1_CT<br>S                  | USART2_TX                                             | SPI2_NSS/<br>I2S2_WS                            | UART4_RX                                     | USART1_CTS<br>/ USART1_NSS | I2C1_SCL                     | FDCAN1_RX                       | -        | -                          | -    | -         | -                      | EVENTOUT |
|        | PA12 | -                 | TIM1_ETR          | -                     | LPUART1_RT<br>S                  | USART2_RX                                             | SPI2_SCK/<br>I2S2_CK                            | UART4_TX                                     | USART1_RTS                 | I2C1_SDA                     | FDCAN1_TX                       | -        | -                          | -    | -         | COMP2_OUT              | EVENTOUT |
|        | PA13 | JTMS/SWDIO        | -                 | -                     | -                                | -                                                     | -                                               | -                                            | -                          | -                            | -                               | -        | -                          | -    | -         | COMP1_OUT              | EVENTOUT |
|        | PA14 | JTCK/SWCLK        | -                 | -                     | -                                | -                                                     | -                                               | -                                            | -                          | -                            | -                               | -        | -                          | -    | -         | -                      | EVENTOUT |
| Port B | PA15 | JTDI              | TIM2_CH1          | TIM1_CH2N             | LPTIM1_ETR                       | I2C1_SMBA                                             | SPI1_NSS/<br>I2S1_WS                            | SPI2_NSS/<br>I2S2_WS                         | USART2_CK                  | UART4_RTS                    | -                               | -        | USART1_TX                  | -    | TIM8_CH4N | TIM2_ETR               | EVENTOUT |
|        | PB0  | -                 | TIM1_CH2N         | -                     | TIM8_CH2N                        | -                                                     | SPI1_MISO/<br>I2S1_SDI                          | -                                            | USART2_TX                  | UART4_CTS                    | -                               | -        | -                          | -    | -         | -                      | EVENTOUT |
|        | PB1  | -                 | TIM1_CH3N         | -                     | TIM8_CH3N                        | -                                                     | SPI2_NSS/<br>I2S2_WS                            | -                                            | USART2_RX                  | -                            | -                               | -        | -                          | -    | -         | COMP1_OUT              | EVENTOUT |
|        | PB2  | RTC_OUT2          | -                 | -                     | TIM8_CH4N                        | SPI1_RDY                                              | LPTIM1_CH1                                      | SPI2_SCK/<br>I2S2_CK                         | SPI2_MOSI/<br>I2S2_SDO     | -                            | -                               | -        | -                          | -    | -         | -                      | EVENTOUT |
|        | PB3  | JTDO/<br>TRACESWO | TIM2_CH2          | -                     | I3C1_SDA                         | I2C1_SDA                                              | SPI1_SCK/<br>I2S1_CK                            | FDCAN2_RX                                    | USART1_CK                  | LPUART1_TX                   | I2C1_SCL                        | CRS_SYNC | USART1_TX                  | -    | TIM8_CH1  | UART5_RTS              | EVENTOUT |
|        | PB4  | NJTRST            | -                 | TIM15_CH1             | I3C1_SCL                         | LPTIM1_CH2                                            | SPI1_MISO/<br>I2S1_SDI                          | FDCAN2_TX                                    | SPI2_NSS/<br>I2S2_WS       | LPUART1_CT<br>S              | I2C1_SDA                        | -        | USART1_RX                  | -    | TIM8_CH2  | UART5_CTS              | EVENTOUT |
|        | PB5  | -                 | -                 | -                     | I3C1_SDA                         | I2C1_SMBA                                             | SPI1_MOSI/<br>I2S1_SDO                          | -                                            | -                          | LPUART1_RT<br>S              | FDCAN2_RX                       | -        | USART1_CK                  | -    | TIM8_CH3  | UART5_RX               | EVENTOUT |
|        | PB6  | -                 | -                 | -                     | I3C1_SCL                         | I2C1_SCL                                              | -                                               | SPI1_MISO/<br>I2S1_SDI                       | USART1_TX                  | LPUART1_TX                   | FDCAN2_TX                       | -        | USART1_CTS<br>/ USART1_NSS | -    | TIM8_CH4  | UART5_TX               | EVENTOUT |

